|INCH-POUND|

MIL-R-83726/13F(USAF)
9 August 1991
SUPERSEDING
MIL-R-83726/13E(USAF)
16 August 1985

MILITARY SPECIFICATION SHEET

RELAYS, SOLID STATE, TIME DELAY (ON OPERATE), TYPE I CLASS C, SPSTNO, 300 MILLIAMPERES, FIXED TIME, 0.05 TO 600 SECONDS, HERMETICALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

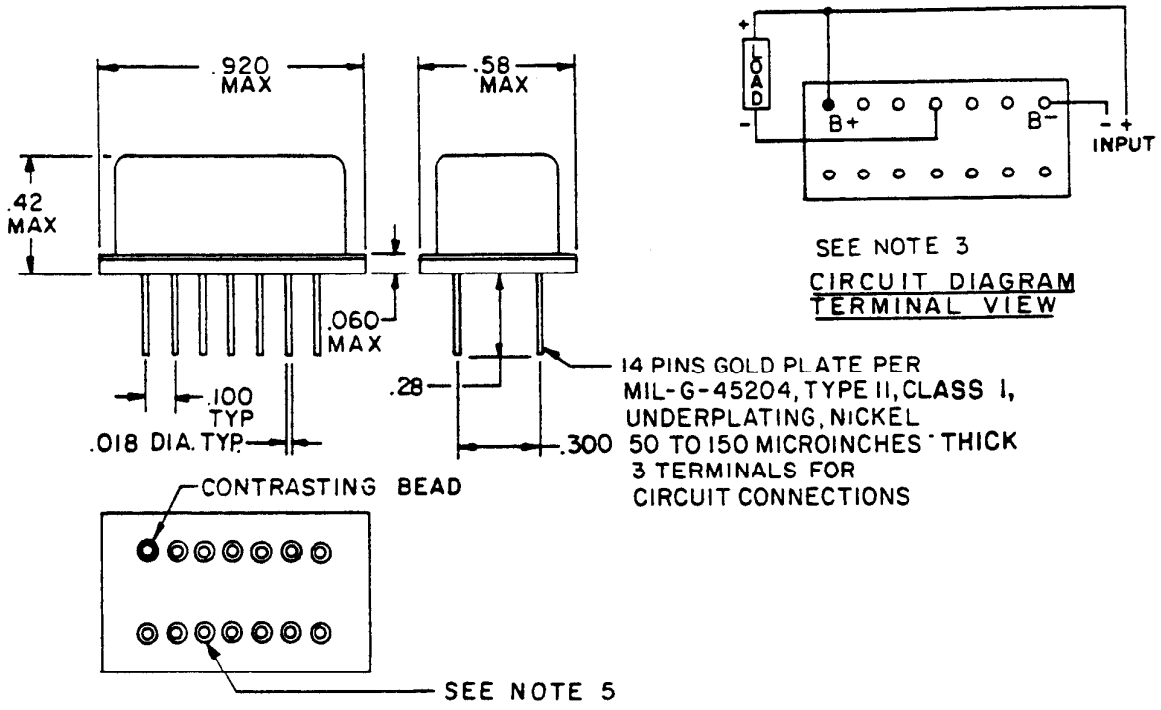
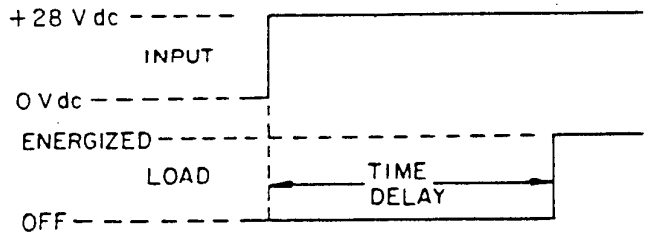


FIGURE 1. Outline dimensions and configuration of relay.

Ⓕ denotes changes

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TIMING DIAGRAM

Inches	mm
.018	0.46
.06	1.5
.100	2.54
.28	7.1
.300	7.62
.42	10.7
.58	14.7
.920	23.37

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Load is connected between B+ and terminal designated LOAD. Delay begins upon application of power to terminals (B+) and (B-). Upon completion of delay period, circuit from load terminal to ground is completed. (B+) and (B-) are shown for reference only; they do not appear on the relay.
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. This is an active pin. Do not connect this pin to any portion of the circuit or ground.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

TABLE I. Available time delay relays. 1/ 2/

Dash number	Time delay seconds ±10%	Dash number	Time delay seconds ±10%
-0500	.05	-1602	16
-1000	.10	-1802	18
-2000	.20	-2002	20
-3000	.30	-2202	22
-5000	.50	-2502	25
-7500	.75	-3002	30
-1001	1.0	-3502	35
-2001	2.0	-4002	40
-3001	3.0	-4502	45
-4001	4.0	-5002	50
-5001	5.0	-5502	55
-6001	6.0	-6002	60
-7001	7.0	-1003	100
-8001	8.0	-2003	200
-9001	9.0	-3003	300
-1002	10	-4003	400
-1202	12	-5003	500
-1402	14	-6003	600

1/ Additional time delay relays within the 0.05 to 600 second delay range are available. To establish Part or Identifying Numbers (PIN's) not listed in table I (see "PIN" herein).
2/ The suffix letter W, X, or Y to designate quality level shall be added to the dash number (see "PIN" herein).

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REQUIREMENTS:

Operational data:

Timing action: Delay-on-operate.

Timing delay: Fixed time, 50 milliseconds to 600 seconds. See table I.

Timing accuracy: ±10 percent of the nominal timing under all conditions of input voltage and environmental extremes.

Recycle characteristics:

Before time out: A power interruption occuring after the start but before completion of the timing cycle shall be for a duration of 0.5 percent of the nominal time delay or 10 ms, whichever is greater, to ensure a loss in timing no greater than 10 percent.

After time out: A power interruption of 0.5 percent of nominal time delay or 10 ms, whichever is greater, will initiate a new timing cycle with a loss in timing no greater than 5 percent.

Input data:

Input voltage: 28 V dc nominal; range 18 V dc to 31 V dc.

Duty rating: Continuous.

Current drain: 10 milliamperes maximum, plus load current.

Reverse polarity protection: The timer will not be damaged or operate when input voltage polarity is reversed.

Output data:

Configuration: SPSTNO, solid state switch closure to ground.

Load ratings:

Resistive: 300 milliamperes at 25°C, derated to 100 milliamperes at +125°C. Minimum load current of 10 milliamperes is required.

Inductive: Three MIL-R-5757/9 relays. (Any relay with 26.5 V dc coil voltage.)

Lamp load: Two MS25237-327 lamps in accordance with MIL-L-6363.

Motor load, dc: Not applicable.

Inductive load, ac: Three MIL-R-39016/6 relays.

Resistive load, ac: Not applicable.

Motor load, ac: Not applicable.

Load suppression: Suppression for inductive loads for output protection is provided within the unit.

Voltage drop: 2.5 volts maximum at -55°C and +25°C, 2.0 volts maximum at +125°C.

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Leakage current: 1 microampere maximum at 25°C, 10 microamperes maximum at +125°C.

Continuous current: Not applicable.

Overload: Not applicable.

Electrical data:

Insulation resistance: 1,000 megohms minimum at 500 V dc. 1/

Dielectric withstanding voltage: 500 V rms, 60 Hz, (sea level). 1/

Transients:

Voltage surge: In accordance with MIL-STD-704, figure 9, limit 1, for category B equipment.

Self-generated spikes: ±10 V.

Spike susceptibility: Not applicable.

Radio interference: Not applicable.

Environmental data:

Temperature range: -55°C to +125°C.

Altitude: 80,000 feet.

Shock: 150 g's for 11 ±1 ms half-sine wave.

Vibration (sinusoidal): 10 Hz to 80 Hz at 0.06 inch DA. 80 Hz to 3,000 Hz at 20 g's.

Vibration scan: Not applicable.

Acceleration: Not applicable.

Seal: MIL-STD-202, method 112, condition C.

Solderability: Not applicable.

Physical data:

Dimensions and configuration: See figure 1.

Terminal strength: Bend test not applicable.

Weight: 12 grams maximum.

(F) Marking: See MIL-R-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

1/ Measured between all terminals tied together to the case.

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Ⓕ ESDS protection program: An ESDS protection program shall be implemented within 6 months of the date of revision F to this document. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

Ⓕ ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

Ⓕ ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

Ⓕ PIN: Consists of the prefix M83726/13-, a four-digit dash number expressed in milliseconds, and a quality level indication as follows:

	M	83726/13-	1001	W
Military designation				
Specification sheet number				
The first three digits are significant; the fourth digit is the number of zeros to follow the first three digits. The time delay is expressed in milliseconds and converted to seconds. (See examples.)				
Quality level indication (W, Y, or X). Refer to MIL-R-83726. 2/				

EXAMPLES: M83726/13-0500W - 50 millisecond time delay, W level
M83726/13-1001X - 1 second time delay, X level
M83726/13-6003Y - 600 second time delay, Y level

2/ Any relays numbered prior to the date of this specification without a quality level indication shall be considered interchangeable (store and issue) with the "W" quality level.

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F713-02)

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